

Width vs Depth in Multi-Layer Perceptrons for Noisy Sketch Classification

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https://github.com/Junaid522/DL_PA2

Abstract—This report investigates the representational and generalization trade-offs between shallow-wide and deep-narrow Multi-Layer Perceptrons (MLPs) on a 15-class subset of the Quick, Draw! dataset. Under strict constraints (no CNNs, $< 3\text{M}$ parameters, ≤ 40 epochs), we analyze width vs depth efficiency, hierarchical representation learning, and generalization behaviour. Experimental results show that deeper architectures provide superior parameter efficiency and improved validation performance compared to purely wide networks. We further analyze confusion patterns and discuss the role of aleatoric ambiguity in sketch classification.

I. INTRODUCTION

The Quick, Draw! dataset presents a challenging classification problem due to noisy, incomplete, and highly variable human sketches. Unlike structured datasets such as MNIST, the data lacks spatial consistency and strong local structure.

The objective is to design high-performing classifiers using only fully connected layers (MLPs), without convolutional inductive biases. The assignment imposes strict constraints:

- No CNNs
- Maximum 3 million parameters
- Maximum 40 epochs

This setting allows investigation of the fundamental trade-off between width and depth in neural networks.

II. DATASET AND PREPROCESSING

Each image is a 28×28 grayscale bitmap flattened to a 784-dimensional vector.

A. Normalization

Input features were standardized using training mean and variance:

$$x_{norm} = \frac{x - \mu}{\sigma}$$

This significantly improved optimization stability.

B. Data Augmentation

Small affine transformations (translation and rotation) were applied to improve invariance and generalization. Since MLPs lack spatial priors, augmentation proved essential.

III. PART A: PANCAKE MODEL (WIDTH FOCUS)

A. Architecture

- 1–2 hidden layers
- 1024–2048 neurons per layer
- GELU activation
- Dropout regularization

B. Results

Wide networks achieved high training accuracy but moderate validation performance.

Observation: Increasing width led to rapid parameter growth with diminishing validation improvement.

Validation Accuracy: $\approx 80.37\%$

C. Analysis

Shallow-wide networks rely heavily on memorization capacity. While Universal Approximation Theorem guarantees expressivity, practical optimization and generalization suffer due to inefficient parameter usage.

IV. PART B: TOWER MODEL (DEPTH FOCUS)

A. Architecture

- 5–8 hidden layers
- 128–256 neurons per layer
- Layer Normalization
- GELU activation

B. Results

Deep architectures outperformed wide networks despite fewer total parameters.

Validation Accuracy: $\approx 77.44\%$

C. Training Challenges

Deeper models initially exhibited unstable gradients. This was mitigated using:

- Layer Normalization
- Careful initialization
- Learning rate scheduling

V. PART C: CHAMPION MODEL

A. Final Architecture

$784 \rightarrow 1024 \rightarrow 512 \rightarrow 256 \rightarrow 128 \rightarrow 15$

- GELU activations
- Layer Normalization
- Dropout (0.15–0.2)
- Weight decay
- Label smoothing
- OneCycle learning rate schedule

Total Parameters: $\approx 1.23\text{M}$ **Best Validation Accuracy:**
 $\approx 80.28\%$

VI. PART D: THEORETICAL ANALYSIS

A. 1. Width vs Depth

The Universal Approximation Theorem states that a single hidden layer can approximate any continuous function given sufficient width.

However, in practice:

- Representing compositional functions requires exponentially many neurons in shallow networks.
- Deep networks reuse intermediate features.
- Hierarchical representations (lines $\rightarrow$ shapes $\rightarrow$ objects) emerge naturally in deep architectures.

Thus, depth improves **parameter efficiency** and **generalization**. While shallow networks can represent the function, deep networks represent it more compactly.

B. Why Deep Architectures are Preferred in Vision

Image tasks contain compositional structure. Even though MLPs lack convolutional bias, deeper layers progressively abstract features:

$\text{Pixels} \rightarrow \text{Edges} \rightarrow \text{Shapes} \rightarrow \text{Object Concepts}$

Depth enables feature reuse and smoother decision boundaries.

C. 2. Confusion Matrix Analysis

The most confused class pairs were:

- Cookie vs Donut
- Apple vs Basketball

These classes share strong visual similarity (circular structure). Inspection suggests that confusion primarily arises from **aleatoric uncertainty** (intrinsic data ambiguity) rather than epistemic underfitting.

Even human annotators may disagree on such sketches.

VII. PART E: ARCHITECTURE RATIONALE

A. Why This Depth and Width?

The champion architecture balances:

- Sufficient depth for hierarchical abstraction
- Moderate width for representational capacity
- Parameter efficiency under 3M constraint

B. Overfitting Mitigation

We employed:

- Data augmentation
- Weight decay
- Dropout
- Label smoothing
- Learning rate scheduling

Training and validation curves showed improved generalization compared to purely wide networks.

C. Model Comparison

Model	Parameters	Epochs	Val Acc
Pancake	5.8M	39	80%
Tower	0.5M	40	77%
Champion	1.2M	29	80%

TABLE I
COMPARISON OF ARCHITECTURES

VIII. CONCLUSION

This study demonstrates that depth provides superior parameter efficiency and generalization compared to naive width scaling in MLPs. Although shallow networks satisfy universal approximation theory, deeper models better capture compositional structure in noisy sketch data. The champion model achieves strong performance within strict constraints, validating the importance of hierarchical representation learning even without convolutional inductive bias.

AI USAGE DECLARATION

Generative AI was used to assist in refining theoretical explanations and report structuring. All model implementation, experimentation, and evaluation were independently conducted.